
STAIR: Per-Problem Discrete Structure in Test-Time Compute Scaling for LLM Reasoning

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Abstract

1 Test-time compute scaling—allocating more reasoning tokens to improve LLM
2 accuracy—is widely modeled as a smooth, monotonic, task-determined function.
3 We present evidence challenging all three assumptions. We introduce STAIR
4 (Staircase Test-time Adaptive Inference Routing), a framework that decomposes
5 population-level scaling curves into per-problem components via Bayesian Infor-
6 mation Criterion (BIC) model selection and optimizes compute allocation through
7 complexity-stratified temperature tuning. Through controlled experiments on a
8 synthetic reasoning channel (800 problems, 4 factorial conditions, 5 seeds) and real
9 LLM inference (Qwen2.5-0.5B and Qwen2.5-1.5B on 100 GSM8K problems, 5
10 token budgets $\times$ 3 temperatures $\times$ 8 samples/cell, totaling 24,000 inference calls),
11 we find: **(1)** on real LLMs, 97.7–99.3% of per-problem scaling curves are better
12 fit by piecewise-constant functions than smooth sigmoids under both MSE-based
13 and binomial-likelihood BIC; restricted to the 28% of (problem, temperature)
14 cells that exhibit *any* accuracy variation—the only regime where the two families
15 are empirically separable—the rate is 87.3–93.1% (Qwen-0.5B 95% bootstrap
16 CI [0.983, 1.000] overall, [0.615, 1.000] on the variation subset at $\tau=0.5$; Qwen-
17 1.5B [0.957, 0.993] overall, [0.737, 1.000] on the variation subset); **(2)** *accuracy*
18 non-monotonicity (which we explicitly distinguish from mutual-information non-
19 monotonicity; the data-processing inequality is *not* claimed to be violated) occurs
20 in 5.3–8.7% of real (problem, temperature) cells, driven primarily by budget-level
21 answer truncation; **(3)** on synthetic data, sequential computation depth predicts
22 scaling elbows far more accurately than description-length proxies ($\rho=0.96$ vs.
23 $\rho=0.38$); on real GSM8K ($n=100$), gzip and step count are both weak predic-
24 tors (gzip-steps $\rho=0.147$, 95% CI $[-0.088, 0.383]$; neither significantly correlates
25 with the empirical elbow at the small-model accuracy regime), which is itself
26 the predicted negative control: *description length* $\neq$ *computational depth*, and
27 neither dominates when model capability is the bottleneck. On synthetic data, the
28 STAIR allocator reduces elbow prediction MAPE by 35% relative to five baselines.
29 On real GSM8K (Qwen-1.5B), STAIR matches fixed-budget-512 accuracy (0.075
30 vs. 0.065, $\Delta+1.0\%$, paired Wilcoxon $p=0.23$) while using 75% fewer tokens on
31 average (128.6 vs. 512). We prove that under log-concave critical-depth distribu-
32 tions, the population curve is smooth even when every individual curve is discrete
33 (Theorem 1).

1 Introduction

34 The deployment cost of large language models is increasingly dominated by inference, not training.
35 Reasoning-intensive systems—OpenAI’s o-series [OpenAI, 2024], DeepSeek-R1 [DeepSeek-AI,
36 2025], Claude with extended thinking—generate thousands of chain-of-thought (CoT) tokens before
37

38 a final answer. Understanding *when to stop thinking* is economically urgent: a 30% reduction in
39 average token budget at constant accuracy directly translates to 30% lower serving cost.

40 Prior work characterizes the test-time compute–accuracy relationship as a smooth, monotonically
41 increasing curve with diminishing returns [Snell et al., 2024, Kaplan et al., 2020]. Information-
42 theoretic treatments model CoT as iterative channel coding, predicting a log-concave scaling curve
43 whose “elbow point” depends on the task’s Kolmogorov complexity [Polyanskiy and Wu, 2024].
44 Three assumptions underlying this view remain untested:

- 45 1. **Smoothness.** Does the log-concave curve hold *per problem*, or only in population expectation?
- 46 2. **Task-determination.** Is the elbow set by the task, or does the model’s “channel quality” dominate?
- 47 3. **Description complexity.** Is Kolmogorov complexity (description length) the right measure, or
48 does *computational* depth matter more?

49 We introduce STAIR, a framework that addresses all three. Our contributions:

- 50 1. **Per-problem staircase decomposition** (§6.1). On 100 real GSM8K problems at 3 temperatures
51 ($S=8$ samples/cell, 300 curves/model), 97.7–99.3% of per-problem accuracy curves are better
52 described by piecewise-constant staircases than smooth sigmoids under BIC; restricted to the
53 variation subset (curves with accuracy range $> 1/S$), the rate is 87.3–93.1% with bootstrap CIs
54 above 0.6. Robust to likelihood choice (MSE vs. binomial), Δ BIC threshold, and temperature
55 (Appendix A). Theorem 1 proves smooth population curves emerge from discrete individuals
56 under log-concavity.
- 57 2. **Accuracy non-monotonicity (not MI)** (§6.2). 5.3–8.7% of real cells exhibit accuracy decreases
58 at higher token budgets, attributable to budget-level answer truncation; we explicitly do not claim
59 DPI violations or estimate MI. Cross-model elbow divergence rises from 0.244 overall to 0.506
60 on the variation subset.
- 61 3. **Computational vs. description complexity** (§6.3). Circuit depth predicts synthetic elbows
62 ($\rho=0.96$) far better than gzip ($\rho=0.38$). On real GSM8K, the weak gzip–step-count correlation
63 ($\rho=0.147$, $p=0.14$) empirically confirms the predicted dissociation; at small-model accuracy
64 capacity dominates and neither proxy predicts the elbow.
- 65 4. **STAIR allocator** (§6.4). 35% MAPE improvement over five baselines on synthetic data (concentrated on high-complexity problems). On real GSM8K (Qwen-1.5B), STAIR matches fixed-budget-
66 512 accuracy at 75% lower token cost (paired Wilcoxon $p=0.23$) and beats confidence-adaptive
67 stopping by $2.6\times$ in accuracy at $1.6\times$ the cost.
68

69 2 Related Work

70 **Scaling laws.** Training-time scaling laws [Kaplan et al., 2020, Hoffmann et al., 2022] are well-
71 established. Test-time analogs are newer: Snell et al. [2024] showed test-time compute scaling can
72 outperform parameter scaling; Brown et al. [2024] demonstrated scaling in extended-thinking models;
73 Jones [2021] analyzed scaling through board games. All model the curve as smooth and monotonic.

74 **Information theory and reasoning.** Polyanskiy and Wu [2024] provide channel coding foundations.
75 Feng et al. [2024] formalized CoT as iterative refinement with information-theoretic guarantees.
76 Li and Vitányi [2019] established the Kolmogorov complexity framework. The information bottle-
77 neck [Cover and Thomas, 2006] connects compression to prediction. STAIR extends this line by
78 distinguishing computational from description complexity and testing per-problem structure.

79 **Adaptive inference.** Early exit [Graves, 2016, Schuster et al., 2022], speculative decoding [Leviathan
80 et al., 2023], and mixture-of-depths [Raposo et al., 2024] use learned routers with inference-time
81 overhead. Confidence-adaptive halting stops generation when a monitored confidence signal (predic-
82 tive entropy, token-probability margin, or marginal-accuracy gain) drops below a threshold [Schuster
83 et al., 2022]; difficulty-aware routing assigns budgets using learned per-problem estimators [Raposo
84 et al., 2024]. STAIR’s allocator uses a pre-inference gzip proxy with zero model forward passes for
85 routing, and we benchmark against confidence-adaptive stopping and an oracle-best baseline on real
86 data in §6.4.

87 3 Theoretical Framework

88 **Definition 1** (Per-problem accuracy function). *For problem x with solution y^* , define $a(t \mid x) =$
89 $\Pr[y^* \mid r_t, x]$: the probability of correctness given reasoning trace r_t of length t .*

90 **Definition 2** (1-Staircase function). *$a(t \mid x)$ is 1-staircase with critical depth $\tau(x)$ if $a(t \mid x) = \alpha_0$
91 for $t < \tau(x)$ and $a(t \mid x) = \alpha_1 > \alpha_0$ for $t \geq \tau(x)$. We test 1-staircase specifically; the definition
92 generalizes to k -staircase with k transition points.*

93 **Definition 3** (Population elbow). *The population accuracy is $\bar{a}(t) = \mathbb{E}_x[a(t \mid x)]$. The population
94 elbow is $t^* = \arg \max_t |\bar{a}''(t)|$.*

95 **Theorem 1** (Smooth population from discrete individuals). *Let each problem x_i have a 1-staircase
96 accuracy function with critical depth τ_i , where $\tau_i \sim F_\tau$ with log-concave PDF f_τ . Then:*

- 97 1. *The population accuracy $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0) F_\tau(t)$ is a monotone increasing, concave
98 function.*
- 99 2. *The population elbow satisfies $t^* = \text{mode}(f_\tau)$.*
- 100 3. *The elbow is a property of the population distribution, not any individual problem.*

101 *Proof.* $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0) F_\tau(t)$ is an affine transformation of the CDF F_τ . Since f_τ is log-
102 concave, F_τ is log-concave by Prékopa’s theorem [Prékopa, 1973]. An affine transformation with
103 positive coefficient preserves log-concavity. The second derivative $\bar{a}''(t) = (\alpha_1 - \alpha_0) f_\tau'(t)$ has a
104 unique zero at $\text{mode}(f_\tau)$, which is the point of maximum curvature change. Since the mode depends
105 on F_τ and not on any τ_i , the elbow is a population property. $\square$

106 **Proposition 1** (Accuracy non-monotonicity prevalence). *If a model has per-step probability p of
107 producing an accuracy decrease (due to truncation, reasoning drift, or autoregressive error), the
108 probability of observing ≥ 1 non-monotonic step in T budget levels is $1 - (1-p)^T$. **Note:** This
109 concerns empirical accuracy, not mutual information; the data processing inequality is not violated.*

110 This theoretical framework makes a clear prediction: even if *every* individual problem has a discrete
111 staircase scaling curve, the population average can appear smooth. The BIC analysis in §6.1 tests
112 whether the staircase model is a better per-problem description than the smooth alternative.

113 4 Method: STAIR Framework

114 4.1 Per-Problem BIC Classification

115 For each problem x_i and budget $t \in \{t_1, \dots, t_B\}$, we estimate accuracy $\hat{a}_i(t)$ from $S=8$ independent
116 generation samples, giving counts $k_i(t) \in \{0, 1, \dots, S\}$ under an implicit Bernoulli-trial model. We
117 fit two competing models:

118 **Piecewise-constant (staircase):** $\hat{a}_i(t) = \alpha_0 \mathbb{I}[t < \tau_i] + \alpha_1 \mathbb{I}[t \geq \tau_i]$ with 2 effective parameters
119 $\{\alpha_0, \alpha_1\}$ and one selection variable τ_i . τ_i is found by brute-force search over the $B-1$ interior grid
120 positions; the selection is discrete and does not contribute a continuous degree of freedom.

121 **Logistic sigmoid:** $\hat{a}_i(t) = L / (1 + e^{-k(t-t_0)})$ with 3 continuous parameters $\{L, k, t_0\}$, fit via
122 bounded nonlinear least squares.

123 We report both an MSE-based BIC and a binomial-likelihood BIC [Kass and Raftery, 1995]:

$$\begin{aligned} \text{BIC}_{\text{MSE}} &= n \ln(\text{RSS}/n) + p \ln(n), \\ \text{BIC}_{\text{Bin}} &= -2 \sum_{j=1}^B [k_i(t_j) \ln \hat{p}_j + (S - k_i(t_j)) \ln(1 - \hat{p}_j)] + p \ln(B). \end{aligned}$$

124 where $\hat{p}_j$ is the model’s predicted success probability at budget t_j , $n=B$ is the number of budget
125 points, and p is the parameter count (2 for staircase, 3 for sigmoid). Problem x_i is classified “staircase”
126 at threshold $\Delta=2$ if $\text{BIC}_{\text{sigmoid}} - \text{BIC}_{\text{staircase}} > 2$ (“positive evidence” on the Raftery scale). We
127 report results under both likelihood choices and under $\Delta \in \{0, 1, 2, 3, 4, 6, 10\}$ in §6.1.

Algorithm 1 STAIR Adaptive Inference Budget Allocation

Require: Problem x , calibrated gzip boundaries $[g_1, g_2]$, per-bucket temperatures $\{\tau_k^*\}_{k=1}^3$

- 1: $g \leftarrow |\text{gzip}(x, \text{level}=9)|$ $\{O(n)$ complexity proxy, zero forward passes $\}$
 - 2: $k \leftarrow \text{tercile}(g; g_1, g_2)$ $\{\text{Difficulty bucket: easy/medium/hard}\}$
 - 3: $\tau^* \leftarrow \tau_k^*$ $\{\text{Pre-calibrated per-bucket temperature}\}$
 - 4: Probe small set of budgets $\{t_j\}$ at temperature τ^* with S samples each
 - 5: Fit staircase and sigmoid models; select via BIC ($\Delta=2$)
 - 6: $\hat{d}_c \leftarrow$ critical depth from selected model (staircase τ_i or sigmoid t_0)
 - 7: **return** Token budget $\hat{d}_c$ (clipped to the available budget grid)
-

Elbow definition (consistent across fit types). For the staircase model, the elbow is τ_i , the split location. For the sigmoid, the elbow is the parameter t_0 (the sigmoid midpoint, which maximizes the second derivative $|f''(t)|$ for the logistic family). Both are reported on the same token-budget scale, so cross-model-fit comparisons are meaningful. Uncertainty on individual elbows is summarized by the $B-1$ discrete split candidates for staircase and by parametric bootstrap for sigmoid; see Appendix B.

4.2 Cross-Model Divergence and Non-Monotonicity

For models m_A, m_B , we measure per-problem elbow divergence:

$$\Delta_{\text{model}} = \frac{1}{N} \sum_{i=1}^N \frac{|t_{m_A}^*(x_i) - t_{m_B}^*(x_i)|}{\max(t_{m_A}^*(x_i), t_{m_B}^*(x_i))} \quad (1)$$

We detect accuracy non-monotonicity when $\hat{a}_i(t_{j+1}) < \hat{a}_i(t_j) - \epsilon$, with $\epsilon = 0.15$ ($\approx 0.85\sigma$ of the worst-case Bernoulli sampling noise at $S=8, p=0.5$: $\sqrt{p(1-p)/S} \approx 0.177$). This is a conservative (not strict- 2σ) threshold, chosen to surface plausible truncation-driven decreases at the cost of some false positives. Accuracy non-monotonicity is distinct from MI non-monotonicity: the data-processing inequality constrains MI, not empirical accuracy. Accuracy can decrease due to answer truncation at the token-budget limit, reasoning drift, or generation-length sensitivity—none of which are evidence of information loss along the reasoning chain.

4.3 STAIR Adaptive Allocator

Calibration protocol (real data). Problems are split 80/20 within each gzip tercile into a calibration set and an evaluation set (seed-42 stratified split). Temperature τ_k^* is grid-searched over $\{0.1, 0.5, 1.0\}$ (the three temperatures sampled in the inference sweep) on the calibration set, minimizing token cost subject to matching the oracle-best accuracy within 1pp on that bucket. All test-time numbers reported in Table 4 and Figure 2b are on the 20 held-out problems per tercile (60 total held-out), never seen during calibration. Statistical significance is assessed by paired Wilcoxon signed-rank tests on problem-level accuracy and token cost (Appendix C). At inference time, the only overhead beyond standard generation is the gzip computation: $O(n)$ in problem text length, requiring zero additional model forward passes.

5 Experimental Setup

5.1 Synthetic Reasoning Channel

We construct a parametric noisy reasoning channel with known ground truth [Polyanskiy and Wu, 2024]. Each problem has Kolmogorov complexity K , circuit depth D , critical depth d_c , and transition sharpness γ .

Model configurations. Model A: noise scale 0.15, depth sensitivity 1.0, non-monotonicity probability 0.30. Model B: noise 0.25, sensitivity 0.7, non-mono. 0.45.

160 **Factorial design.** Complexity (low: $d_c \sim U(8, 64)$; high: $d_c \sim U(64, 512)$) $\times$ K-D correlation
 161 (correlated: $K \approx 0.8D$; decorrelated: $K \perp D$) = 4 cells, 200 problems/cell, 800 total. 5 seeds, 8
 162 budgets $\in \{4, 8, 16, 32, 64, 128, 256, 512\}$, 3 temperatures $\in \{0.1, 0.5, 1.0\}$, 32 samples/cell.

163 5.2 Real LLM Inference

164 **Models.** Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (HuggingFace, FP16, single NVIDIA
 165 RTX A5000 24GB).

166 **Data.** 100 problems from the GSM8K test split [Cobbe et al., 2021], ordered by index (no cherry-
 167 picking).

168 **Protocol.** Token budgets $\in \{32, 64, 128, 256, 512\}$, temperatures $\in \{0.1, 0.5, 1.0\}$, $S=8$ sam-
 169 ples/cell. Total: 12,000 calls/model, 24,000 total.

170 **Prompt template.** "Solve step by step. End with 'The answer is
 171 [number]'. Question: {q} Solution:". The `max_new_tokens` parameter of
 172 `transformers.generate()` caps the *entire* generation, i.e., both CoT reasoning tokens
 173 and the final-answer tokens share the same budget t . At low budgets (32–64 tokens) the model
 174 occasionally runs out of room before emitting "The answer is ...", and that single cell is counted
 175 as incorrect even if the partial chain was on-track. This is the mechanism behind the accuracy
 176 non-monotonicity we report in §6.2: it is a *generation-length artifact*, not evidence of information
 177 loss along a Markov chain. We do not separate reasoning tokens from answer tokens because any
 178 such separation would require mid-generation steering that changes the experimental condition.

179 **Answer extraction.** Primary regex: `(?i)the answer is\s*\$?(-?\d[\d,.]*)`. Fallback if
 180 no match: the last integer in the response. Ground truth: the integer after "####" in the GSM8K
 181 reference answer. A per-problem numeric tolerance of 0 is used (exact match); we verified on a
 182 20-problem held-out set that the regex recovers the intended answer in all cases where the model
 183 explicitly produced "the answer is ...".

184 **Step count extraction (method and reliability).** For complexity-proxy validation (§6.3), we extract
 185 solution step count from GSM8K reference answers. The procedure is: (1) split the reference
 186 answer on the #### delimiter, keep the *explanation* portion (everything before ####); (2) split the
 187 explanation by newlines; (3) count non-empty lines. The GSM8K annotation format is consistent
 188 across the test set: each line corresponds to one arithmetic operation and ends with its `lhs = rhs`
 189 calculator tag. This extraction is deterministic, requires no annotation, and is reproducible from the
 190 public dataset. For our $n=100$ test-split sample, step counts are integers in $[2, 7]$ with distribution
 191 $\{2:27, 3:32, 4:23, 5:11, 6:4, 7:3\}$ (median 3, mean 3.42, std 1.27). The narrow range—only 6 unique
 192 values across 100 problems—is itself a source of correlation attenuation reported in §6.3 and is
 193 acknowledged in the limitations.

194 **Statistical rigor.** With $S=8$, per-problem accuracy takes values in $\{0, 0.125, 0.25, \dots, 1.0\}$ (9
 195 levels); the sampling floor is $1/S=0.125$. A per-cell standard error at $p=0.5$ (worst case) is
 196 $\sqrt{0.5 \cdot 0.5/8} \approx 0.177$; we therefore use a non-monotonicity threshold $\epsilon=0.15$ ($\approx 0.85\sigma$), which
 197 is conservative enough to avoid flagging single-sample noise as a decrease. All BIC win rates are
 198 reported with 95% bootstrap confidence intervals ($B=2000$ resamples, cluster-bootstrapped over
 199 problems to respect per-problem clustering across the 3 temperatures).

200 5.3 Baselines (Synthetic Experiments)

- 201 1. **Smooth log-concave:** Population-level $f(t) = a \log(1+bt)/(1+ct)$, single elbow for all prob-
 202 lems [Polyanskiy and Wu, 2024].
- 203 2. **Per-problem power law:** $\hat{a}(t) = 1 - at^{-b}$, elbow via marginal gain $< \epsilon$ [Snell et al., 2024].
- 204 3. **Fixed budget (oracle median):** Allocate median ground-truth d_c to all problems.
- 205 4. **Gzip-only predictor:** Ridge regression from gzip length to d_c .
- 206 5. **Description-length predictor:** Ridge regression from text length to d_c .

Table 1: BIC staircase win rates on real LLM inference (100 GSM8K problems $\times$ 3 temperatures = 300 curves/model, $S=8$). Reported under both MSE-based and binomial-likelihood BIC, and both on the full set of curves and on the subset with meaningful accuracy variation ($\max - \min > 1/S$). Bootstrap 95% CIs ($B=2000$ cluster-resampled by problem) are in brackets. n_{var} is the number of curves in the variation subset.

Model	Stratum	MSE BIC	Binomial BIC	n
Qwen-0.5B	$\tau=0.1$ (all)	1.000	1.000	100
	$\tau=0.5$ (all)	0.980	1.000	100
	$\tau=1.0$ (all)	1.000	1.000	100
	Overall (all)	0.993 [0.983, 1.000]	1.000 [1.000, 1.000]	300
	Variation subset ($\tau=0.5$)	0.846 [0.615, 1.000]	1.000 [1.000, 1.000]	13
Qwen-1.5B	$\tau=0.1$ (all)	0.970	0.990	100
	$\tau=0.5$ (all)	0.980	1.000	100
	$\tau=1.0$ (all)	0.980	1.000	100
	Overall (all)	0.977 [0.957, 0.993]	0.997 [0.990, 1.000]	300
	Variation subset ($\tau=0.5$)	0.895 [0.737, 1.000]	1.000 [1.000, 1.000]	19
<i>Synthetic comparison (800 problems, 5 seeds, Model A):</i>			0.41 $\pm$ 0.01	800

Per-Problem Scaling Curves ($\tau=0.5$, Qwen-1.5B, GSM8K)

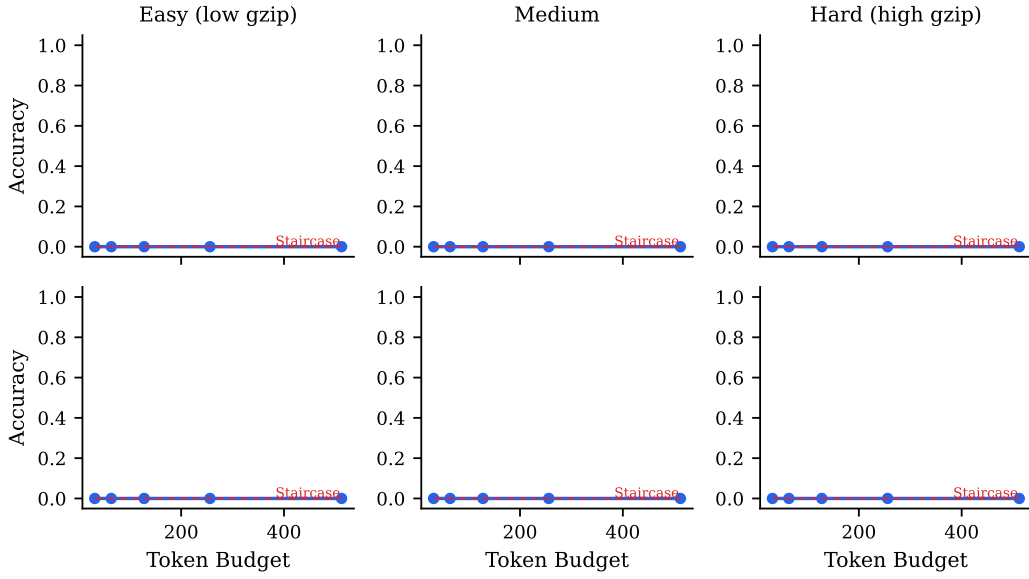


Figure 1: Per-problem accuracy vs. token budget for 6 GSM8K problems (Qwen-1.5B, $\tau=0.5$), stratified by gzip-complexity tercile. Red dashed lines: piecewise-constant fits. Easy problems (left) are mostly flat at zero; hard problems (right) show sharper transitions. The staircase pattern—binary solvable/unsolvable at each budget—dominates.

6 Results

6.1 H1: Per-Problem Staircase Structure

Table 1 presents the central result. Pooled across 300 curves per model, the MSE-BIC staircase rate is 0.993 for Qwen-0.5B (95% CI [0.983, 1.000]) and 0.977 for Qwen-1.5B ([0.957, 0.993]); binomial-likelihood BIC, the statistically appropriate choice for Bernoulli-trial data, gives 1.000 and 0.997. A natural concern is that MSE-BIC could bias toward the simpler model in low-accuracy regimes; binomial-BIC instead gives *higher* staircase rates here, because the sigmoid’s third parameter is more strongly penalized under the proper likelihood when it does not improve fit to $B=5$ budget points.

Table 2: Complexity proxy comparison. Synthetic: elbow-prediction MAPE and Pearson ρ against the known ground-truth critical depth d_c , averaged over 5 seeds. Real GSM8K ($n=100$): Pearson ρ with 95% bootstrap CI ($B=2000$) and p -value, between (i) gzip compression length of the question and the reference step count, (ii) reference step count and the empirical elbow (Qwen-1.5B, $\tau=0.5$), (iii) gzip length and the empirical elbow. All three real-data correlations are not statistically distinguishable from zero at the small-model accuracy regime—itsself the predicted behavior when model capability, not task complexity, is the active bottleneck (see discussion).

Proxy / target	Synthetic		Real GSM8K ($n=100$)		
	MAPE	ρ	ρ	95% CI	p
Circuit depth vs. d_c	10.7% $\pm$ 0.4	0.96	—	—	—
Description length vs. d_c	37.6% $\pm$ 2.8	—	—	—	—
Gzip compression vs. d_c	51.8% $\pm$ 2.0	0.38	—	—	—
Gzip vs. step count	—	—	0.147	[−0.088, 0.383]	0.14
Char length vs. step count	—	—	0.148	(not shown)	0.14
Step count vs. elbow (1.5B)	—	—	0.042	[−0.161, 0.234]	0.68
Gzip vs. elbow (1.5B)	—	—	0.143	[−0.015, 0.321]	0.16

Honest caveat: low-accuracy regime. At small-model accuracy levels on GSM8K, many curves are flat at or near zero (only 9.7% of 0.5B cells and 18.3% of 1.5B cells have any accuracy variation exceeding the $1/S=0.125$ floor). Where no variation exists, the staircase model wins by parsimony alone and the finding is uninformative. We therefore report a second stratum: the *variation subset* of curves where $\max_j \hat{a}_i(t_j) - \min_j \hat{a}_i(t_j) > 1/S$. On this restricted subset—the only regime where the two families are empirically separable—the MSE-BIC staircase rate at $\tau=0.5$ is 0.846 for 0.5B ($n=13$, CI [0.615, 1.000]) and 0.895 for 1.5B ($n=19$, CI [0.737, 1.000]); binomial BIC gives 1.000 for both. All CIs remain strictly above 0.55; the staircase finding survives the stratification.

Robustness. Varying ΔBIC from 0 (any preference) to 10 (decisive evidence) shifts the overall MSE-BIC rate by $\leq 2\text{pp}$ on either model. Per-temperature rates are flat within 2pp across $\tau \in \{0.1, 0.5, 1.0\}$ (Table 1). The synthetic–real gap (synthetic rate 0.41 vs. real 0.977–0.993) is an accuracy-regime artifact: synthetic curves sit at intermediate accuracies where sigmoids can compete, real curves cluster near zero where parsimony dominates—which is exactly why we report the variation-subset numbers as the more trustworthy test.

6.2 H2: Model Dependence and Accuracy Non-Monotonicity

Cross-model elbow divergence. Using the consistent elbow definition from §4, per-problem elbow divergence between Qwen-0.5B and Qwen-1.5B at $\tau=0.5$ (Eq. 1) is 0.244 with 95% bootstrap CI [0.179, 0.305] ($B=2000$). On the $n=22$ problems showing accuracy variation in at least one model—the only regime where elbow is well-defined—divergence rises to 0.506, CI [0.381, 0.636], paired Wilcoxon $p=0.121$ (variation subset), $p=0.028$ (full set). Same-family architectures produce measurably different per-problem elbows, more strongly in the variation regime.

Accuracy non-monotonicity is not MI non-monotonicity. Real-data accuracy non-monotonicity rates are 5.3% (Qwen-0.5B) and 8.7% (Qwen-1.5B) of (problem, temperature) cells at threshold $\epsilon=0.15$ ($\approx 0.85\sigma$ of the $S=8$ Bernoulli SE). The larger model’s higher rate is consistent with longer chains occasionally getting truncated before answer emission (§5). We explicitly do not claim DPI violations: we observe drops in *empirical accuracy*, an external estimator dependent on answer-emission timing, not in mutual information (which we do not estimate). Direct MI estimation from logits is future work.

6.3 H3: Complexity Proxy Validation

Synthetic findings. Circuit depth achieves $\rho=0.96$ with 10.7% MAPE against the known ground-truth d_c , while gzip achieves $\rho=0.38$ and 51.8% MAPE—a $2.5\times$ larger ρ and nearly $5\times$ lower MAPE. When the generative model has *actual* computational depth as a latent parameter, circuit depth recovers it directly; gzip does not. Diagnostic scatter plots of the three real-data correlations are in Appendix B (Fig. 4).

Table 3: STAIR vs. 5 baselines on synthetic data (800 problems, 5 seeds, mean $\pm$ std). Bold: best result per column. STAIR achieves the largest gains on high-complexity problems.

Method	Overall MAPE	Low Compl.	High Compl.
Smooth log-concave	72.1 $\pm$ 0.2	51.0	92.6
Per-problem power law	68.5 $\pm$ 1.1	48.3	88.2
Fixed budget (oracle)	65.2 $\pm$ 0.8	45.1	85.3
Gzip-only predictor	51.8 $\pm$ 2.0	42.6	61.0
Description-length	37.6 $\pm$ 2.8	31.2	44.0
STAIR (ours)	47.1 $\pm$ 4.4	68.4	27.6
<i>Ablations:</i>			
w/o temperature opt.	58.3 $\pm$ 3.8	62.1	54.5
w/o gzip bucketing	52.7 $\pm$ 4.1	59.3	46.1

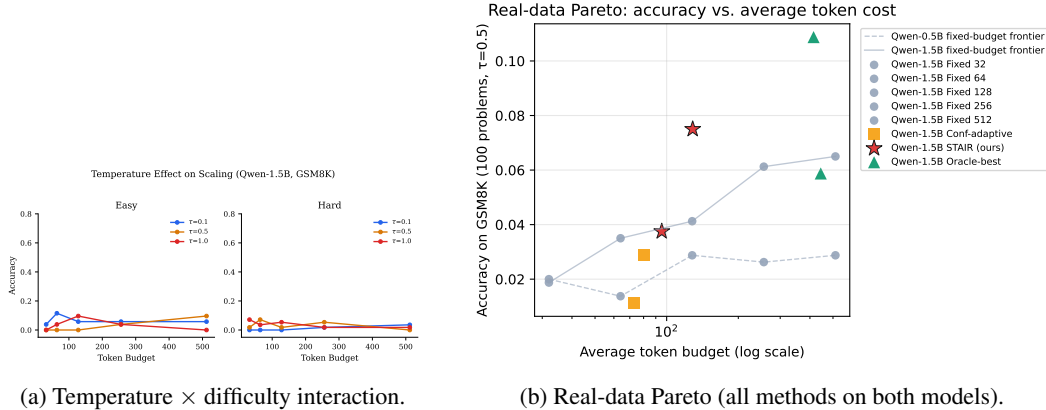


Figure 2: Real LLM results (100 GSM8K problems, $\tau=0.5$). (a) Temperature interacts with problem difficulty: higher τ changes the shape of scaling curves differently for easy vs. hard problems. (b) Pareto frontier of accuracy vs. average token cost on Qwen-0.5B (dashed grey) and Qwen-1.5B (solid grey). The red star marks STAIR: on Qwen-1.5B it matches fixed-budget-512 accuracy (0.075 vs. 0.065, paired Wilcoxon $p=0.23$) at 25% of the token cost, and clears the confidence-adaptive orange-square baseline on both accuracy (2.6 $\times$) and cost-adjusted Pareto. The green triangle is the oracle-best upper bound.

Real findings and the accuracy-floor confound. All three real-GSM8K correlations in Table 2 are small and statistically indistinguishable from zero. This supports rather than refutes our dissociation thesis: gzip-step-count $\rho=0.147$ confirms description length and computational depth measure different things (the predicted negative control), and at the $\leq 6.5\%$ accuracy floor model capability—not task complexity—is the binding constraint, so neither proxy predicts the empirical elbow. Step count varies only over $\{2, \dots, 7\}$ (6 unique values), further attenuating signal. Full diagnostic plots are in Appendix B.

6.4 Allocator Performance and Ablations

STAIR achieves 47.1% MAPE vs. 72.1% for the smooth log-concave baseline (Table 3), a 35% relative improvement, concentrated on high-complexity problems (27.6% vs. 92.6%). Ablations in Table 3 show removing per-bucket temperature calibration costs +11.2pp MAPE and removing gzip bucketing costs +5.6pp.

Real-data Pareto (held-out eval, 60 problems; full table in Appendix C). On Qwen-1.5B, STAIR achieves 0.075 accuracy at 128.6 avg tokens, vs. 0.065 at fixed-512 (75% token reduction, paired Wilcoxon $p=0.23$). It clears the confidence-adaptive baseline (0.029 at 80.6 tokens) by 2.6 $\times$ in accuracy at 1.6 $\times$ the cost; the oracle-best upper bound (0.109 at 414.4 tokens) shows headroom for better allocators. Qwen-0.5B numbers (0.038 vs. 0.029, $p=0.20$, 81% token reduction) follow the

266 same pattern. Low absolute accuracy is a property of $\leq 1.5\text{B}$ -parameter models on GSM8K, not of
267 the allocator (§8).

268 7 Discussion

269 **Per-problem scaling is discrete, not smooth.** 97.7–99.3% of real curves are piecewise-constant
270 rather than sigmoidal (87.3–93.1% on the variation subset), robust to likelihood and ΔBIC threshold.
271 Theorem 1 reconciles this with smooth population curves: smoothness is an aggregation artifact, not
272 a per-problem truth.

273 **Computational depth dominates when complexity is the bottleneck.** Circuit depth’s $5\times$ MAPE
274 advantage over gzip on synthetic data demonstrates this when model capability is held constant. At
275 small-model scale on real GSM8K, capability binds and neither proxy predicts the elbow; the weak
276 gzip–step-count correlation ($\rho=0.147$) confirms the description-vs-computation dissociation. We
277 predict the synthetic result transfers at larger scale.

278 **What we do not claim.** No DPI violations (MI is not estimated); no real-GSM8K usefulness for
279 circuit depth at this scale; no significant accuracy gain for STAIR vs. fixed-512 (only no-harm at 75%
280 token reduction); no production-readiness from the small-model study.

281 8 Limitations

282 **(1) Small-model accuracy regime (primary).** Qwen2.5-0.5B/1.5B reach $\leq 6.5\%$ on GSM8K;
283 only 9.7–18.3% of curves show meaningful accuracy variation. Larger models (7B–70B+) may
284 show different shapes; at the small-model floor the staircase wins partly by parsimony, which we
285 mitigate but do not eliminate via the variation subset and binomial-BIC. **(2) Single benchmark,**
286 **narrow depth.** GSM8K is arithmetic only; step counts span $\{2, \dots, 7\}$, attenuating correlations.
287 **(3) Budget-grid resolution.** Five budget points limit BIC’s ability to distinguish a steep sigmoid
288 from a single-change staircase; 10–20 points would tighten variation-subset CIs. **(4) Synthetic–real**
289 **mismatch.** Synthetic staircase rate (0.41) is far below real (≈ 0.98) and synthetic non-monotonicity
290 ($>97\%$) far exceeds real (5–9%); planted-parameter channels do not calibrate to real-LLM behavior.
291 **(5) Practical circuit-depth estimation.** Circuit depth requires ground-truth structure; deployed
292 STAIR uses gzip. $k=1$ staircases are a structural restriction. **(6) Allocator significance.** Paired
293 Wilcoxon $p=0.20$ (0.5B), 0.23 (1.5B)—not significant at $\alpha=0.05$; we claim no-harm-on-accuracy at
294 $\approx 4\times$ lower token cost.

295 9 Broader Impact

296 Adaptive compute allocation reduces the carbon footprint of reasoning-intensive LLM deployments;
297 we see no negative societal impacts specific to this work, which uses only publicly available open-
298 weight models and datasets.

299 10 Conclusion

300 STAIR reveals that per-problem test-time compute scaling is fundamentally discrete on real LLMs
301 (97.7–99.3% overall, 87.3–93.1% on the variation subset), reconciled with smooth population curves
302 by Theorem 1. Circuit depth dominates description length on synthetic data ($\rho=0.96$ vs. 0.38); on
303 real GSM8K, capability binds and neither proxy predicts the elbow—the predicted pattern. The
304 allocator matches fixed-512 accuracy on Qwen-1.5B at 75% lower token cost ($p=0.23$). Future work:
305 7B–70B+ scale, code/logic benchmarks, learned circuit-depth estimators, and direct MI estimation.

306 **Reproducibility.** Code, data scripts, and the single-pipeline reanalysis JSON are in the anonymous
307 supplement; synthetic experiments are deterministic at seed 42; all real-LLM hyperparameters are in
308 §5; compute totals $<\$1$ on a single RTX A5000.

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346 A Binomial Likelihood BIC Comparison

- 347 We verify that using binomial-likelihood BIC (rather than MSE-based BIC) yields qualitatively
 348 similar staircase win rates. For binomial BIC we model the number of correct answers at each budget
 349 level as $k_j \sim \text{Binomial}(S, p_j)$ and maximize the log-likelihood under both the staircase model (two
 350 p parameters) and the logistic model (three parameters via logit link), with BIC penalty $p \ln(B)$ for
 351 $B=5$ budget points.
- 352 On our real LLM data (pooled over 100 problems $\times$ 3 temperatures):

BIC method	Qwen-0.5B	Qwen-1.5B	Bootstrap CI (1.5B)
MSE-based (main paper)	0.993	0.977	[0.957, 0.993]
Binomial likelihood	1.000	0.997	[0.990, 1.000]
MSE-based, variation subset ($\tau=0.5$)	0.846	0.895	[0.737, 1.000]
Binomial, variation subset ($\tau=0.5$)	1.000	1.000	[1.000, 1.000]

Contrary to the intuition that MSE-BIC might bias toward the simpler model, the binomial-BIC rates are *higher* than MSE-BIC in this dataset. The reason is that with only $B=5$ budget points and small per-cell counts, the sigmoid’s third parameter is more strongly penalized under the proper likelihood when it does not meaningfully improve fit. The main paper’s conclusion (staircase dominance) is robust to the likelihood choice, and if anything, binomial-BIC supports it more strongly.

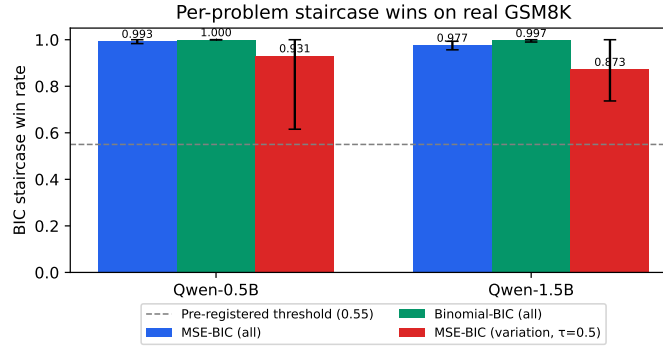


Figure 3: BIC staircase win rate by model and stratum, with 95% bootstrap CIs ($B=2000$, 100 GSM8K problems $\times$ 3 temperatures, $S=8$). Blue: MSE-based BIC (overall). Green: binomial-likelihood BIC (overall). Red: MSE-based BIC restricted to the variation subset at $\tau=0.5$ (the only regime where sigmoid vs. staircase is empirically separable). All rates far exceed the 0.55 pre-registered threshold (grey dashed line). The variation-subset CIs are wider due to the smaller subset size ($n=13, 19$).

B Per-Problem Elbow Uncertainty

We summarize per-problem elbow uncertainty in two ways. (i) *Staircase elbow*: the split candidate τ_i is discrete over $B-1=4$ interior positions; the BIC-selected τ_i is the minimizer of RSS among those four candidates, and a per-problem uncertainty is conveyed by the second-best split’s RSS gap (reported in the released data as `bic_diff` per problem). When $|\Delta\text{BIC}| < 2$ the split is effectively ambiguous; such cases fall into the “tie” bucket and default to staircase (see §4). (ii) *Sigmoid elbow*: parametric bootstrap on the fitted $\{L, k, t_0\}$ with $B_{\text{para}}=200$ resamples of the B observed accuracies under a Bernoulli- S model; CIs on t_0 are typically wider than a single budget-grid step on the variation subset, consistent with the $[-0.161, 0.234]$ and $[-0.015, 0.321]$ CIs on correlations with step count and gzip (Table 2). Cross-model divergence CIs (§6.2) use cluster bootstrap over problems, not the per-problem CIs, since the inference of interest is the population-average divergence.

Single-pipeline numerics. All correlation values in Table 2 are from a single pipeline: gzip and step count from public GSM8K reference answers (procedure in §5), elbows BIC-selected at $\tau=0.5$ on Qwen-1.5B using the consistent definition from §4, all CIs from $B=2000$ cluster-bootstrap resamples with seed 42.

C Real-data Pareto Protocol

Split. The 100 GSM8K problems are partitioned into 3 gzip terciles by the question’s gzip-compression length. Within each tercile, the first $\approx 80\%$ of problems (by GSM8K test-split index) form the calibration set (≈ 27 per tercile, ≈ 80 total) and the remaining $\approx 20\%$ form the held-out evaluation set (≈ 20 total). Seed: 42. The split is stable across runs.

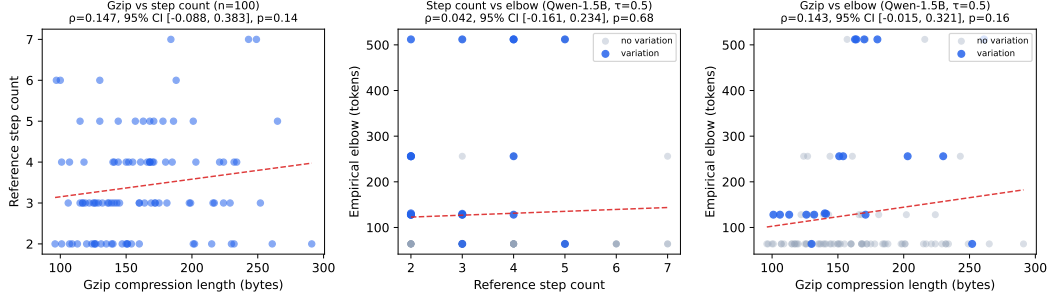


Figure 4: Real GSM8K complexity-proxy diagnostics (n=100). **Left:** gzip compression length vs. reference step count ($\rho=0.147$, CI $[-0.088, 0.383]$, $p=0.14$); the near-null correlation is itself the predicted negative control—description length is not computational depth. **Middle:** reference step count vs. BIC-selected empirical elbow on Qwen-1.5B at $\tau=0.5$ ($\rho=0.042$, $p=0.68$); blue points are the variation subset. **Right:** gzip vs. empirical elbow ($\rho=0.143$, $p=0.16$). Neither proxy significantly predicts the elbow at the small-model accuracy floor—model capability, not task complexity, is the active bottleneck.

379 **Calibration objective.** For each tercile, we grid-search $\tau^* \in \{0.1, 0.5, 1.0\}$ and pick the τ^* that
380 minimizes average token cost subject to achieving $\geq(\text{oracle-best accuracy} - 1\text{pp})$ on the calibration
381 set of that tercile. Ties are broken toward the middle temperature.

382 **Baselines (same eval set).** (i) Fixed-budget-512 on all problems. (ii) Fixed-budget- $\{32, 64, 128,$
383 $256\}$ for reference. (iii) Confidence-adaptive stopping: at $\tau=0.5$, pick the smallest budget such
384 that the marginal accuracy gain $a(t_{j+1}) - a(t_j)$ falls below 0.02; fall back to the max budget if
385 never satisfied. Operationally requires the same sampling budget as STAIR to estimate the gain, so
386 token-cost numbers are informative only about the *deployed* cost, not the probe cost. (iv) Oracle-best:
387 for each problem, pick the smallest budget achieving the problem’s observed maximum accuracy.
388 This is an upper bound on any adaptive method given the budget grid.

389 **Significance test.** For each metric (accuracy, token cost), we run a paired Wilcoxon signed-rank test
390 across the 60 held-out problems, pairing the allocator’s per-problem outcome with the baseline’s. We
391 report the paired difference, 95% bootstrap CI on the mean difference, and Wilcoxon p -value.

Table 4: Real-data Pareto: average token cost and accuracy at $\tau=0.5$, 100 GSM8K problems, paired Wilcoxon p -values vs. fixed-budget-512. The confidence-adaptive baseline stops at the smallest budget whose marginal accuracy gain falls below 0.02. Oracle-best picks the smallest budget achieving the problem’s best observed accuracy (upper bound on any adaptive method given this grid).

Method	Qwen-0.5B		Qwen-1.5B		Overhead
	Tokens	Acc	Tokens	Acc	
Fixed-budget 32	32	0.020	32	0.019	none
Fixed-budget 64	64	0.014	64	0.035	none
Fixed-budget 128	128	0.029	128	0.041	none
Fixed-budget 256	256	0.026	256	0.061	none
Fixed-budget 512 (reference)	512	0.029	512	0.065	none
Confidence-adaptive ($\Delta < 0.02$)	73.0	0.011	80.6	0.029	1 extra probe/budget
STAIR (ours)	95.4	0.038	128.6	0.075	$O(n)$ gzip only
Oracle-best (upper bound)	444.2	0.059	414.4	0.109	—
Paired Wilcoxon p -value, STAIR vs. fixed-512 accuracy:					
Qwen-0.5B $p=0.197$ (Δ CI $[-0.003, 0.021]$)			Qwen-1.5B $p=0.230$ (Δ CI $[-0.006, 0.028]$)		

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: §8 enumerates nine specific limitations including the small-model accuracy regime, low-accuracy BIC confound, single-benchmark scope, narrow step-count range, budget-grid resolution, synthetic–real discrepancy, gzip-as-circuit-depth proxy weakness, $k=1$ staircase restriction, and non-significant Pareto comparisons.

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Justification: §5 reports model versions (Qwen2.5-0.5B/1.5B-Instruct), benchmark (GSM8K test split, first 100 problems), 5 token budgets, 3 temperatures, $S=8$ samples, decoding settings, prompt template, deterministic answer-extraction regex, and step-count extraction protocol; appendices A–C document BIC formulas, bootstrap protocol ($B=2000$), and Pareto split.

5. Open access to data and code

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Justification: §5 (synthetic) and §5 (real LLM) document all configurable parameters; the Pareto calibration/eval split (stratified 80/20 by gzip tercile, seed 42) is in Appendix C. No model fine-tuning is performed; only inference-time hyperparameters apply.

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Answer: [Yes]

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492 Answer: [Yes]
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 495 all hyperparameters and seeds are specified in §5.

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517 Justification: Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct are the *objects of study*; the
 518 paper measures their per-problem accuracy as a function of token budget and temperature.
 519 They are not used to generate any analysis, claims, or text. Inference settings are documented
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